



Pramaan

Proving a document is real — in seconds.

AI document-authenticity and fraud detection for government schemes. Mathematics decides what mathematics can decide — and the system says plainly what it could not verify.

THE PROBLEM

Aadhaar proves who you are. It does not prove what your paper says.

Most welfare fraud in India is not a stolen identity. It is a real, correctly-authenticated person submitting a document that says something untrue about them.

Income certificate

says ₹96,500 — the bank statement says otherwise

Caste / domicile certificate

carries a seal the issuing officer never applied

Bank guarantee

backs a tender, and the bank has never heard of it

Checked how? One clerk, hundreds of applications, no forensic tooling, no time. A competent forgery passes. An honest applicant with an odd-looking document gets rejected. Both failures are invisible.

Leakage is measured, acknowledged, funded against

₹3.48L cr

cumulative DBT savings from removing leakage

5.2 cr

fake ration cards purged

830/1,572

minority institutions found fake or inactive

₹202 cr

MP Jal Nigam fake bank-guarantee case

We deliberately do not overclaim this. DBT's savings came mostly from Aadhaar *identity* deduplication — removing ghost people — not document forensics. Pramaan targets the complementary gap: the document is fake even though the person is real. No consolidated "MP loses ₹X crore to fake documents" figure exists publicly, and we will not invent one.

THE SOLUTION

Upload a document. Get a defensible verdict.

- **Verdict** — Authentic / Suspicious / Likely fake, with a 0–100 risk score
- **Red flags** — each quoting the exact evidence on the document
- **Cross-field logic** — does the arithmetic and the story hold up?
- **Identifier validation** — Aadhaar, PAN, GSTIN, IFSC, IBAN, MRZ checksums
- **12-module forensic dossier** — printable, auditable
- **What it could NOT verify** — stated explicitly, never hidden

SAMPLE VERDICT

LIKELY FAKE risk 88

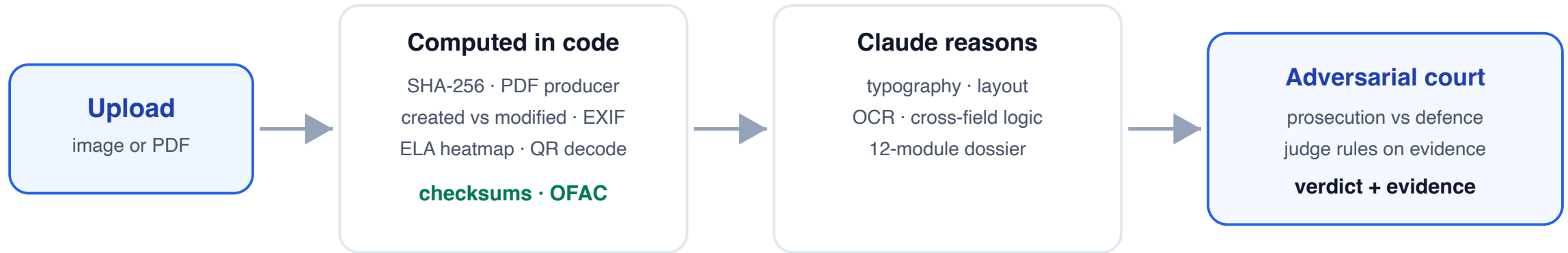
Income figure re-typed — baseline drift + weight mismatch over a wiped region

Serial altered — localised compression anomaly

Declared income contradicts bank balance

Identifier fails checksum — mathematically impossible to have been issued

Deterministic facts first. AI reasoning second.



The order is the product. Anything a computer can prove, a computer proves — and the AI is not permitted to argue with it.

We made the AI argue with itself

A single AI pass either cries fraud at every compression artifact, or waves everything through.

Prosecution

Builds the strongest honest case that it is forged
— citing only observable evidence.

Defence

Gives the innocent explanation — scanner artifacts, recompression, a legitimate re-save — and exposes overreach.

Judge

Weights both against **binding** code-computed facts. Records what was **decisive** — and what it **dismissed**.

That dismissal list is the false-positive guard — and the officer sees it. A failed checksum cannot be argued away; an OFAC list that failed to download cannot be spun into "clean".

What we actually tested

CLAIM	EVIDENCE
Identifier validation is real mathematics	60/60 unit tests pass — Verhoeff, PAN, GSTIN mod-36, IFSC, IBAN mod-97, ICAO 9303 MRZ
Forensic pipeline runs with no API key	<code>npm run selftest</code> → 5/5 pass
ELA responds to real pixel edits	mean error 0.196 → 0.264 when a splice was introduced
Watchlist screening is live, not mocked	real OFAC SDN export — 19,393 records parsed

And what we refuse to claim: Error-Level Analysis is weak on flat, vector-rendered documents — ordinary body text produces as much error as a splice. We report it as one weak indicative signal, never as a verdict. We would rather tell you that than be caught by a judge who knows it.

What we can connect to — and what it really costs

SOURCE	WHAT IT PROVES	REALITY
Aadhaar offline QR / XML	Signed identity data vs UIDAI public key	Buildable now
OFAC / UN / EU lists	Watchlist screening	Live today
ICAO 9303 MRZ	Passport / ID check digits	Shipped
PAN / GSTIN / MCA / IEC	Registry existence and status	Paid aggregator
DigiLocker / e-District	Did this office actually issue it?	Needs partnership

DigiLocker is the endgame — it turns *"internally consistent"* into *"issued by that office."* It needs a government sponsor, which is exactly what a department pilot unlocks.

One scheme. One department. Real documents.

A pilot gives us the two things we cannot generate ourselves: a labelled dataset to calibrate confidence against, and the sponsorship needed for DigiLocker issuer verification.

What we measure

- Fraudulent documents caught that manual review missed
- False positives — honest applicants wrongly flagged
- Officer time per application
- Turnaround, submission to decision

Where it sits

- Counter-level triage — clerk uploads, gets a reason
- Audit trail — every verdict stores its evidence
- **Officer decides.** Pramaan recommends; it never auto-rejects

github.com/Yashraj00700/pramaan